

ABRA — the plain-English deck

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A slide-by-slide, jargon-light tour. The white paper (linked on the last slide) has the math and sources.

Slide 1 — ABRA

A family of small AI models for competitive Pokémon (Champions VGC). Built from thousands of real ladder games. Honest about what it can and can't know.

Slide 2 — The big idea

You **can't reliably predict who wins** a match from the two team sheets — in this format it's close to a coin flip, and even a strong player-rating model barely beats a coin. So ABRA stops trying to call the winner and instead **helps you make better decisions**: what to bring, and what to do turn by turn.

This is the same move sports analytics made with "expected goals" — you can't predict the final score, but you can measure the value of each shot. ABRA is that idea, pointed at Pokémon.

Slide 3 — How it's built

One rule: **keep every real game, and analyse on top of it**. ABRA saves every public ladder replay into a growing library and builds its models on that library — so it gets smarter as more games come in, and never has to re-download anything. It runs on a normal laptop; no special hardware.

Slide 4 — The town of models

Each model is a "house" you can visit on the site:

- **MEDICHAM** — the damage engine. Matches the community standard, but disagrees with the game's OFFICIAL engine by a wide margin, so it is being replaced by that engine and kept only as a lookup.
 - **MEW** — plays the official engine against itself, so we are not limited to the games people happen to upload.
 - **GURU** — reads the metagame: which team styles beat which, from real results, with error bars.
 - **XATU** — reads the opponent: the likely item, ability, and moves behind each Pokémon.
 - **PORY** — mid-battle win chance. **Retracted as a finding**: it works out to counting how many Pokémon each side has left and how healthy they are, and it does not beat exactly that baseline.
 - **The scoring bot** — the part that finally *looks at the board*.
 - **SLOWKING** — the strategist: there's no single best team, so it plays a smart mix.
 - **CHOMP** — the team picker: which four to bring, which two to lead.
 - **ALAKAZAM** — the coach (in progress): the best move to make, right now.
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Slide 5 — The win: the practice opponent can finally see

Everything ABRA claims about "this build beats that build" comes from playing the game out against a practice opponent. That opponent used to pick its move purely by **how popular the move is** — it had no idea what was standing in front of it. So it fired attacks at Pokémon that were flat immune, it used moves that couldn't possibly work, and when there were two enemies to aim at, it picked one by flipping a coin.

We fixed it by **watching what people actually do**. We took about 48,000 real decisions from games where both players' full teams were public — so we could see not just what someone clicked, but everything they *could* have clicked — and learned how much a real player's choice depends on what's in front of them. Nobody wrote the rules; they were measured.

The result, checked on games the learning never saw:

- it now hits the enemy's weakness **half again as often** as before,
- it wastes **a third fewer** moves on things that simply cannot work,
- and it almost never fires at something immune any more.

It is still short of a human on all three, and one thing it still does badly is decide **when to switch** — that's the next job. But the practice opponent is no longer playing blind, which means the numbers it produces are worth more than they were.

PORY also beats a coin flip mid-game, using how many Pokémon each side has left and their health, and it's *calibrated* — when it says 70% it's really about 70%. It's live on the site as a per-turn "you're at X%".

Slide 6 — The honest parts (this is a feature)

Two things we tested and reported straight, even though they're negatives:

- **Picking the team doesn't beat a coin.** We tested whether the team-picker's choices track who wins — they don't, more than chance. So we don't oversell it. The damage math is still exact and useful.
- **The "rock-paper-scissors" metagame is a hint, not a fact.** The data suggests Trick Room beats Hyper Offense beats Sand beats Trick Room — but each of those rests on only ~13–18 games, so the error bars are wide. We label it suggestive, and it'll firm up as more games arrive.

Saying what we *can't* prove, as plainly as what we can, is the whole point.

Slide 7 — What's next

- **Teach it when to switch.** The scoring bot chooses its attacks well now, but it still decides when to swap Pokémon at random — and swapping is a big part of how the game is really played. It's the same machinery, pointed at one more decision.
- **Re-run the build experiments.** Every "this build is better" result on record was measured against the *old*, blind opponent. They all need re-measuring before they can be trusted.
- **ALAKAZAM** — the in-battle coach. A light version runs in your browser; a version that could beat top humans needs a rented cloud computer and months of the AI playing itself to learn.
- **A million games.** Real games trickle in ~18k/week; the AI playing itself can generate millions fast, which is what the coach needs to get truly strong.

Slide 8 — Read more

Full technical detail, math, and sources: [ABRA white paper](#). Also: [project summary](#) · [technical docs](#) · [model ledger](#).